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Marine vessels and convertible windshield apparatuses for marine vessels

Abstract

A marine vessel has a hull with a bow, a stern, starboard and port sides, a deck on the hull, and a console located on the deck between the starboard side and the port side and between the bow and the stern. The console has a helm for the marine vessel and a console windshield which shields the helm from wind. A cover extends over the helm. An auxiliary windshield is coupled to the cover, the auxiliary windshield being movable into and between a stowed position in which the auxiliary windshield is stowed in the cover and a deployed position in which the auxiliary windshield extends downwardly from the cover into a gap between the console and one of the starboard side and the port side.

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Background/Summary

FIELD

(1) The present invention relates to marine vessels and to convertible windshield apparatuses for marine vessels.

BACKGROUND

(2) The following U.S. Patents are incorporated herein by reference.

(3) U.S. Pat. No. 7,144,061 discloses a retractable windshield and collapsible console apparatus which can be raised and the windshield extended when needed to afford additional protection to persons from conditions due to environment, and to allow for convenient and safe stowing of the apparatus in the same location without causing an obstruction. The console can be collapsed and stowed into a position which is substantially parallel relative to a base on the floor or other surface of a boat, other vehicle or other structure when not in use.

(4) U.S. Pat. No. 9,688,363 discloses an improved walkthrough door for a marine vessel comprising a curved roof with a windshield disposed in the curved roof proximate a foredeck of the marine vessel. A transparent walkthrough door is slidably mounted in the curved roof and moveable between a closed position, whereat the walkthrough door seals with the foredeck and an open position permitting access to the foredeck. Preferably, the walkthrough door is electrically operated.

(5) U.S. Pat. No. 10,800,494 discloses an exemplary windshield assembly for a marine vessel including a plurality of first extrusions arranged vertically; a plurality of second extrusions,

wherein each second extrusion is a crossbar arranged between a pair of first extrusions; and an opening formed between each pair of second extrusions connected between a common pair of first extrusions, wherein each opening includes a plurality of receivers formed for framing a panel, the plurality of receivers formed on a surface of each first extrusion and each second extrusion of a respective opening, and wherein a front opening includes an upper second extrusion and movable front panel, the movable front panel being connected to rotate about the upper second extrusion for venting.

SUMMARY

(6) This Summary is provided to introduce a selection of concepts which are further described below in the Detailed Description. This Summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used as an aid in limiting the scope of the claimed subject matter.

(7) A marine vessel has a hull with a bow, a stern, starboard and port sides, a deck on the hull, and a console located on the deck between the starboard side and the port side and between the bow and the stern. The console has a helm for the marine vessel and a console windshield which shields the helm from wind. A cover extends over the helm. An auxiliary windshield is coupled to the cover, the auxiliary windshield being movable into and between a stowed position in which the auxiliary windshield is stowed in the cover and a deployed position in which the auxiliary windshield extends downwardly from the cover into a gap between the console and one of the starboard side and the port side.

(8) In non-limiting examples, the auxiliary windshield is pivotably coupled to the cover. In the stowed position, the auxiliary windshield is nested in a recess in the cover. In the deployed position the auxiliary windshield and the console windshield together provide a weathertight barrier for the console and the gap.

(9) In non-limiting examples, an actuator is configured to move the auxiliary windshield into the stowed position and into the deployed position. In the deployed position the auxiliary windshield spans an upper portion of the gap.

(10) In non-limiting examples, a wind door is movable into and between a retracted position and an extended position in which the wind door spans a lower portion of the gap. the wind door is pivotably coupled to one of the hull and the console. Moving the auxiliary windshield into the deployed position and moving the wind door into the extended position provides a weathertight barrier for the helm and the gap between the cover and the deck.

(11) In non-limiting examples, the auxiliary windshield is one of starboard and port auxiliary windshields which in the deployed position and together with the console windshield provide a weathertight barrier extending from the starboard side to the port side. The gap is one of starboard and port gaps, and the starboard and port auxiliary windshields span an upper portion of the starboard and port gaps, respectively. Starboard and port wind doors are movable into and between a retracted position and an extended position in which the starboard and port wind doors span a lower portion of the starboard and port gaps, respectively, so that together the console, the starboard and port auxiliary windshields, and the starboard and port wind doors provide a full width weathertight barrier extending from the starboard side to the port side and from the deck to the cover.

(12) In non-limiting examples, a pivot assembly pivotably couples an upper end of the auxiliary windshield to the cover along a pivot axis. The pivot assembly has an upper actuator which pivots the auxiliary windshield into the stowed position and into the deployed position. The upper actuator for example can be a linear actuator coupled to the auxiliary windshield by a rocker arm.

(13) In non-limiting examples, the pivot assembly further has a pivot bracket which is fixedly coupled to the auxiliary windshield. Extension and retraction of the linear actuator pivots the pivot bracket and auxiliary windshield about the pivot axis. The cover comprises a window frame and the pivot bracket comprises a curved outer surface which remains adjacent to the window frame as the

auxiliary windshield is pivoted about the pivot axis into and between the stowed position and the deployed position. A wiper seal on the window frame, wherein the curved outer surface slides along the wiper seal as the auxiliary windshield is pivoted about the pivot axis.

(14) In non-limiting examples, a lower actuator couples a lower end of the auxiliary windshield to the console and to the hull in the deployed position. The lower actuator comprises a linear actuator which is retained in the lower end of the auxiliary windshield, the lower actuator being movable into an active position which engages the lower end of the auxiliary windshield with the console and with the hull, and an inactive position which disengages the lower end of the auxiliary windshield from the console and the hull.

(15) In non-limiting examples, a convertible windshield apparatus for a marine vessel comprises a cover having a recess, an auxiliary windshield being movable into and between a stowed position in which the auxiliary windshield is nested in the recess and a deployed position in which the auxiliary windshield is at least partially out of the recess, and a pivot assembly which pivotably couples the auxiliary windshield to the cover along a pivot axis. The pivot assembly comprises an actuator and a pivot bracket which is fixedly coupled to the auxiliary windshield, wherein the actuator pivots the pivot bracket and auxiliary windshield about the pivot axis.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure includes the following Figures.

(2) FIG. 1 is a perspective view of a marine vessel having a convertible, full width weathertight windshield apparatus.

(3) FIG. 2 is a front view of the marine vessel illustrated in FIG. 1.

(4) FIG. 3 is a starboard side perspective view of a cover and an auxiliary windshield of the windshield apparatus in a deployed position.

(5) FIG. 4 is another starboard side perspective view of the cover and the auxiliary windshield and a wind door in the deployed position.

(6) FIG. 5 is a perspective view looking down at a pivot assembly and the auxiliary windshield and the wind door in the deployed position.

(7) FIG. 6 is a perspective view looking down at the pivot assembly and the auxiliary windshield in the stowed position and the wind door in the deployed position.

(8) FIG. 7 is an exploded view of a cover body, the pivot assembly, and the auxiliary windshield.

(9) FIG. 8 is a sectional view of a lower end of the auxiliary windshield, illustrating a lower actuator.

(10) FIG. 9 is a sectional view of the lower actuator in an active position.

(11) FIG. 10 is a sectional view of the lower actuator in an inactive position.

DETAILED DESCRIPTION

(12) FIGS. 1-2 illustrate a marine vessel 12 comprising a hull 16 having a bow 20, a stern 18, a starboard side 26 and a port side 28. The marine vessel 12 extends from the bow 20 to the stern 18 in a longitudinal direction X, from the starboard side 26 to the port side 28 in a lateral direction Y which is perpendicular to the longitudinal direction X, and from a top to a bottom in a vertical direction Z which is perpendicular to the longitudinal direction X and lateral direction Y.

(13) The marine vessel 12 further comprises a deck 24 (FIG. 6), a console 14 on the deck 24, and a cover 34. The console 14 is located laterally between the starboard side 26 and the port side 28 and longitudinally between the bow 20 and the stern 18. The cover 34 is positioned over the top of the console 14 and extends laterally from the starboard side 26 to the port side 28, and longitudinally over the top of the helm provided at the console 14. Port and starboard gaps 35 or walkways are defined laterally between the console 14 and the starboard side 26 and the port side 28,

respectively, and extend vertically from the deck **24** to the cover **34**. The gaps **35** provide walkways for a passenger to walk back and forth on the deck **24** from the bow **20** to the stern **18**, on both sides of the console **14**.

(14) The console **14** provides the helm at which an operator can control operation of the marine vessel **12**. A console windshield **32** wraps around the console **14** on top of the sidewalls **48** of the console **14**, opening toward the stern **18** and thus shielding passengers located at the helm from wind and other environmental conditions. Support pillars **50** extend generally vertically upwardly and provide support for the panes of the console windshield **32**. The support pillars **50** taper laterally inwardly toward the center of the marine vessel **12**, as illustrated in FIG. 2.

(15) During research and experimentation, the present inventors have realized a need for improved weather barriers for marine vessels having a center console. Marine vessels having center consoles typically lack a full width windshield extending from the starboard side **26** to the port side **28**. In such arrangements, it is known to provide “walk-around functionality” about the center console however providing this functionality typically hinders the ability to provide a full vessel beam weather protection to occupants located at the rear of the vessel. As such, typically only occupants seated at the helm directly behind the center console windshield are protected via the console windshield **32** from the elements. The present inventors have realized a desirability of providing improved apparatuses having convertible, automatically deployable auxiliary windows, for example which hinge downwardly from the vessel's cover into a deployed position. This maintains the walk-around functionality while alternately efficiently providing a full width weathertight barrier which completely extends from the starboard side **26** to the port side **28**. The present inventors have also realized a desirability of providing such auxiliary windshields which are advantageously configured to nest entirely within the cover **34** when stowed, providing both aesthetically pleasing and aerodynamic advantages.

(16) Referring to FIGS. 1-2, the marine vessel **12** includes novel port and starboard auxiliary windshields **40**, which in the illustrated example are pivotably coupled to the cover **34** on the starboard and port sides **26**, **28**, respectively, each being configured to move into and between a stowed position (FIG. 6) and a deployed position (FIGS. 1-5). FIGS. 3-4 are starboard side views of the cover **34** and the starboard side auxiliary windshield **40**. The starboard side of the cover **34** and the starboard side auxiliary windshield **40** are mirror images of the port side of the cover **34** and port side auxiliary windshield **40**. As such, the following description of the starboard side of the cover **34** and starboard side auxiliary windshield **40** equally applies to the port side of the cover **34** and port side auxiliary windshield **40**.

(17) Referring now to FIGS. 3-4, among other things, the cover **34** has an actuator housing **44** and a recess **42** located adjacent to the actuator housing **44**. As best viewed in FIGS. 5-7, the actuator housing **44** has an upper face **108**, a lower cover plate **110**, and a first and second side wall **112a**, **112b**. The actuator housing **44** houses a pivot assembly **72** which is fixedly coupled to the auxiliary windshield **40** and closes the open end **96** of the actuator housing **44**. As further described herein below, the pivot assembly **72** is configured to pivot the auxiliary windshield **40** into and between the stowed and the deployed positions, and the recess **42** is sized and shaped to efficiently store the auxiliary windshield **40** in the stowed position.

(18) The auxiliary windshield **40** has an upper end **52**, a lower end **54**, an outer side **56**, and an inner side **58**. The upper end **52** is pivotably coupled to the pivot assembly **72**. The inner side **58** laterally tapers towards the center of the marine vessel **12** as the auxiliary windshield **40** vertically extends from the lower end **54** to the upper end **52**. The taper of the inner side **58** follows the taper of the console windshield **32**. In the stowed position, the auxiliary windshield **40** is nested within the recess **42** of the cover **34**. In a deployed position, the auxiliary windshield **40** extends downwardly from the recess **42** into an upper portion of the gap **35** between the console **14** and the hull **16**, advantageously providing a weathertight barrier together with the console windshield **32**.

(19) Referring to FIGS. 5-7, the pivot assembly **72** has an upper actuator **74** which is coupled to a

pivot bracket **62** via a rocker arm **76**. In the illustrated examples, the upper actuator **74** is a linear actuator having a body **60** containing an electric motor and an actuation arm **82**. Operation of the electric motor in a first direction extends the actuator arm **82** from the body **60** and operation of the electric motor in an opposite second direction retracts the actuator arm **82** into the body **60**. The type and configuration of the upper actuator **74** can vary and can include any suitable conventional electric, hydraulic, or electro-hydraulic actuator. Suitable actuators are available for purchase from Linak. The body **60** of the upper actuator **74** is pivotably coupled to the upper face **108** of the actuator housing **44** via a boss **80**. The actuation arm **82** is pivotably coupled to the rocker arm **76** via a heim joint **86**. The rocker arm **76** is a generally triangular plate, having three coupling vertices **78a**, **78b**, **78c**. The rocker arm **76** is coupled to the actuator housing **44** at the first coupling vertex **78a** via a boss **84**, coupled to the actuation arm **82** via the heim joint **86** at the second coupling vertex **78b**, and coupled to the pivot bracket **62** at the third coupling vertex **78c** via a linkage arm **88**. The linkage arm **88** is coupled at a first end to the third coupling vertex **78c** via a heim joint **90a** and coupled at an opposite second end to the pivot bracket **62** via a heim joint **90b**.

(20) The pivot bracket **62** has an elongated body **104**, a curved outer surface **92**, first and second ends **106a**, **106b**, and upper and lower halves **114a**, **114b**, which in the illustrated example are formed as one piece. The first and second ends **106a**, **106b** have an aperture **94a**, **94b**, respectively. The curved outer surface **92** is a semi-conical protrusion extending from the elongated body **104** such that the curved outer surface **92** radially expands from the first end **106a** to the second end **106b**, as best illustrated in FIG. 7. The first aperture **94a** is located on the first end **106a** on the upper half **114a**, and second aperture **94b** is located on the second end **106b** on the lower half **114b**, defining a diagonal pivot axis **100** (FIG. 5) about which the pivot bracket **62** rotates.

(21) Referring to FIG. 7, the pivot assembly **72** is pivotably coupled to the actuator housing **44** via a window frame **98** defining the open end **96** of the actuator housing **44**. The auxiliary windshield **40** is coupled to the pivot assembly **72** and the actuator housing **44** via the pivot bracket **62**. The upper end **52** of the auxiliary windshield **40** aligns with and is fixedly coupled to the elongated body **104** along the diagonal pivot axis **100**, such that the auxiliary windshield **40** is pivotable about the diagonal pivot axis **100**. The window frame **98** is positioned on top of the cover plate **110** of the actuator housing **44**, and configured such that the pivot bracket **62** can be stowed and pivoted within a cross sectional area **A** defined by the window frame **98**.

(22) The window frame **98** has a first side **116a**, a second side **116b**, and an upper length **116c**. Each of the first and second sides **116a**, **116b** includes an aperture **118a**, **118b**, respectively. The first side **116a**, the upper length **116c**, and the second side **116b** align with the first side wall **112a**, the upper face **108**, and the second side wall **112b** of the actuator housing **44**. The aperture **118a** of the window frame **98** is aligned with the first aperture **94a** of the pivot bracket **62** and the aperture **118b** of the window frame **98** is aligned with the second aperture **94b** of the pivot bracket **62**. The window frame **98** is secured to the pivot bracket **62** via a first and second pin **120a**, **120b**, such that the first pin **120a** extends through the aperture **118a**, then the first aperture **94a** and the second pin **120b** extends through the aperture **118b** then the second aperture **94b**. The pins **120a**, **120b** are secured to the respective apertures via bearings or bushings and are aligned with one another along the diagonal pivot axis **100**. In some examples, the window frame **98** further includes a wiper seal **22** which extends over and across the length of the elongated body **104** and down the second end **106b** between the window frame **98** and the pivot bracket **62** to provide a sliding watertight seal along the curved outer surface **92** as the auxiliary windshield **40** is pivoted about the diagonal pivot axis **100**.

(23) Referring to FIGS. 8-10, the auxiliary windshield **40** further includes an elongated cavity **36** within the lower end **54** which is concealed by a cover plate **190** (FIG. 3). Additionally, the hull **16** and the console **14** each include an aperture **160a**, **160b** positioned adjacent opposing sides of the lower end **54** when the auxiliary windshield **40** is deployed.

(24) The elongated cavity **36** is configured to store a lower actuator **124** which is coupled on

opposing ends to a first and a second pin **156a**, **156b**. In the illustrated example, the lower actuator **124** is a linear micro-actuator, although this configuration is not limiting. The lower actuator **124** and the first and second pins **156a**, **156b** are configured to couple the lower end **54** of the auxiliary windshield **40** to the console **14** and to the hull **16**. The first and second pin **156a**, **156b** are capable of being extended or retracted laterally from the outer side **56** and the inner side **58** of the auxiliary windshield **40**, into the hull **16** and the console **14**, respectively. The elongated cavity **36** includes first and second dividers **126a**, **126b** which divide the cavity **36** into three sections: an outer section **132a**, a center section **132c**, and an inner section **132b**. The first and the second sections **132a**, **132b** each have an aperture **128a**, **128b**, through which the first and the second pin **156a**, **156b** are configured to extend. The aperture **128a** extends from the outer side **56** of the lower end **54** to the first divider **126a**. The aperture **128b** extends from the inner side **58** of the lower end **54** to the second divider **126b**.

(25) Each of the pins **156a**, **156b** have a first end **134a**, **134b**, a second end **136a**, **136b** and a snap ring **130a**, **130b**. The first end **134a** of the first pin **156a** is stowed within the outer section **132a** of the elongated cavity **36** and configured to extend and retract laterally from the outer side **56** of the auxiliary windshield **40** into the aperture **160a** of the hull **16**. The first end **134b** of the second pin **156b** is stowed within the inner section **132b** of the elongated cavity **36** and configured to extend and retract laterally from the inner side **58** of the auxiliary windshield **40** into the aperture **160b** of the console **14**. The snap ring **130a** is secured around the first pin **156a** within the outer section **132a** of the elongated cavity **36**, and the snap ring **130b** is secured around the second pin **156b** within the inner section **132b**. The second end **136a** of the first pin **156a** and the second end **136b** of the second pin **156b** are coupled to opposing ends of the lower actuator **124** within the center section **132c** of the elongated cavity **36**.

(26) As described above, and best illustrated in FIGS. 5-6, the auxiliary windshield **40** is movable between the deployed position (FIG. 5) and the stowed position (FIG. 6). The upper actuator **74** can be extended or retracted to move the auxiliary windshield **40** into and between the stowed and the deployed position via the pivot assembly **72**. The lower actuator **124** is moveable into an active position which engages the lower end of the auxiliary windshield **40** with the console and with the hull, and an inactive position (FIG. 10) which disengages the lower end **54** of the auxiliary windshield **40** from the console **14** and the hull **16**.

(27) The upper actuator **74** and lower actuator **124** are controlled by one or more user input devices **200**, illustrated in FIGS. 5-6. The user input device **200** is configured to present the user with the ability to deploy or retract at least one of the port and starboard side auxiliary windshields **40**. The user input device **200** can include manually operable electro-mechanical switches or a touch screen located within the vessel **10** and communicatively connected by wired or wireless link to the upper actuator **74** and lower actuator **124**. In one example, the user input device **200** includes an upper actuator switch **210** and a lower actuator switch **220**, each movable into a first and second position. Actuation of the upper actuator switch **210** into the first or the second position causes the upper actuator **74** to extend or retract, pivoting the auxiliary windshield **40** into the stowed or deployed position, respectively. Actuation of the lower actuator switch **220** into the first or second position causes the lower actuator **124** to extend or retract, moving the lower actuator **124** into the active or inactive position, respectively. In an alternative example, the user input device **200** has one main switch, which is movable into a first and a second position. Actuation of the main switch into the first position causes the upper actuator **74** to retract and move the auxiliary windshield **40** into the deployed position, and then causes the lower actuator **124** to extend, moving the lower actuator **124** into the active position. Actuation of the main switch into the second position causes the lower actuator **124** to retract and move the lower actuator into the inactive position, and then extends the upper actuator **74**, moving the auxiliary windshield **40** into the stowed position. Alternative user input devices can be used, including those which allow a user to move the auxiliary windshield **40** into a position between a stowed and deployed position.

(28) In a stowed position, as illustrated in FIG. 6, the upper actuator **74** is extended, such that the actuation arm **82** extends from the body **60** and exerts a rotational force onto the rocker arm **76** via the heim joint **86**. The rotational force holds the linkage arm **88** in place against the pivot bracket **62**, such that the elongated body **104** closes the open end **96** of the actuator housing **44** and the auxiliary windshield **40** is retained within the recess **42**. Optionally, in this position, the lower actuator **124** is moved in the active position, wherein that the pins **156a**, **156b** are extended from the elongated cavity **36** and into engagement with the cover **34** along sides of the recess **42** so as to retain the auxiliary windshield **40** in the stowed position. In this embodiment, the lower actuator **124** is moved from the active position to the inactive position to facilitate subsequent movement of the upper actuator **74** into the retracted position so as to deploy the auxiliary windshield **40**, as described below.

(29) In use, to deploy the auxiliary windshield **40**, the user moves the upper actuator switch **210** into the first position, which causes the lower actuator **124** to move into the inactive position and the upper actuator **74** to retract the actuation arm **82** toward the body **60**, which exerts a pulling force on the rocker arm **76** via the heim joint **86**. The rocker arm **76** is rotated about the boss **84** in a clockwise direction, which exerts a pulling force on the pivot bracket **62** via the linkage arm **88**. The pivot bracket **62** is rotated clockwise about the diagonal pivot axis **100**, which causes the auxiliary windshield **40** to pivot into the deployed position while the curved outer surface **92** remains adjacent to the window frame **98**. The wiper seal **22** is positioned so that the curved outer surface **92** slides along the wiper seal **22** as the auxiliary windshield is pivoted, maintaining a weathertight barrier. When deployed, the auxiliary windshield **40** is angled longitudinally toward the stern **18**, such that the lower end **55** is positioned closer to the bow **20**, as illustrated in FIGS. 1-2. To move the auxiliary windshield **40** from the deployed position into the stowed position, the user moves the upper actuator switch **210** into the second position, which causes the upper actuator **74** to extend the actuation arm **82**, which exerts a pushing force on the rocker arm **76** via the heim joint **86**. The rocker arm **76** is rotated about the boss **84** in a counterclockwise direction, which exerts a pushing force on the pivot bracket **62** via the linkage arm **88**. The pivot bracket **62** is rotated counterclockwise about the diagonal pivot axis **100**, which causes the auxiliary windshield **40** to pivot into the stowed position while the curved outer surface **92** remains adjacent to the window frame **98**. Thereafter the lower actuator **124** is moved into the active position to retain the auxiliary windshield **40** in the stowed position, as described above.

(30) In the inactive position, as illustrated in FIG. 10, the lower actuator **124** is held in place via the pins **156a**, **156b** and the respective snap rings **130a**, **130b** abutting the respective dividers **126a**, **126b**. Extension and retraction of the lower actuator **124** results in movement of the pins **156a**, **156b** asynchronously, insofar as the lower actuator **124** is held in suspension solely by said pins **156a**, **156b**. Upon initiation, the lower actuator **124** exerts an outward force on one of the pins **156a**, **156b**, until the respective snap ring **130a**, **130b** prevents further movement of said pin. The lower actuator **124** continues to extend, exerting a force in the opposite direction and moving the remaining pin into its intended position. To move into the active position, as illustrated in FIG. 9, the user moves the lower actuation switch **220** into the first position, which causes the lower actuator **124** to extend, which forces the pins **156a**, **156b** outward until the respective first ends **134a**, **134b** are positioned within the respective apertures **160a**, **160b** and the respective snap rings **130a**, **130b** abut opposite ends of the elongated cavity **36**. To move into the inactive position, as illustrated in FIG. 10, the user moves the lower actuation switch **220** into the second position, which causes the lower actuator **124** to retract, which exerts a pulling force and moves the pins **156a**, **156b** inward until the respective snap rings **130a**, **130b** abut the respective dividers **126a**, **126b** such that the pins **156a**, **156b** are stowed fully within the lower end **54** of the auxiliary windshield **40**.

(31) Furthermore, the wind doors **64** are movable into and between a retracted positioned and an extended position in which the wind doors **64** span the lower portion of the port and starboard gaps

35. As illustrated in FIGS. 1-2, the auxiliary windshields 40 in the deployed position in combination with the wind doors 64 in the extended position provides a weathertight barrier for the helm and the gaps 35 extending from the starboard side 26 to the port side 28 and from the deck 24 to the cover 34. In the illustrated example, the wind doors 64 are manually pivoted into and between the retracted position and extended position, however it is presently contemplated that an actuator such as an electric motor could be provided for automatically moving the wind doors 64 based on a user input via for example the noted user input device 200.

(32) This written description uses examples to disclose the invention, including the best mode, and to enable any person skilled in the art to make and use the invention. Certain terms have been used for brevity, clarity and understanding. No unnecessary limitations are to be inferred therefrom beyond the requirement of the prior art because such terms are used for descriptive purposes only and are intended to be broadly construed. The patentable scope of the invention is defined by the claims, and may include other examples which occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have features or structural elements which do not differ from the literal language of the claims, or if they include equivalent features or structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. A marine vessel comprising: a hull having a starboard side and a port side; a console having a helm and a console windshield that shields the helm from wind; a cover extending over the helm; and an auxiliary windshield coupled to the cover, the auxiliary windshield being pivotable relative to the cover into a stowed position in which the auxiliary windshield is stowed in the cover and into a deployed position in which the auxiliary windshield extends downwardly from the cover into a gap between the console and one of the starboard side and the port side.
2. The marine vessel according to claim 1, wherein in the stowed position the auxiliary windshield is nested in the cover.
3. The marine vessel according to claim 1, wherein in the deployed position the auxiliary windshield and the console windshield together provide a weathertight barrier across the console and the gap between the console and the one of the starboard side and the port side.
4. The marine vessel according to claim 1, wherein in the stowed position the auxiliary windshield is nested in a recess in the cover.
5. The marine vessel according to claim 1, further comprising an actuator configured to move the auxiliary windshield into the stowed position and into the deployed position.
6. A marine vessel comprising: a hull having a starboard side and a port side; a console having a helm and a console windshield that shields the helm from wind; a cover extending over the helm; an auxiliary windshield coupled to the cover, the auxiliary windshield being movable relative to the cover into a stowed position in which the auxiliary windshield is stowed in the cover and into a deployed position in which the auxiliary windshield extends downwardly from the cover into a gap between the console and one of the starboard side and the port side, wherein in the deployed position the auxiliary windshield spans an upper portion of the gap; and a wind door that is movable into and between a retracted position and an extended position in which the wind door spans a lower portion of the gap.
7. The marine vessel according to claim 6, wherein the wind door is pivotably coupled to one of the hull and the console.
8. The marine vessel according to claim 6, wherein moving the auxiliary windshield into the deployed position and moving the wind door into the extended position provides a weathertight barrier across the gap between the console, the cover, and a deck on the hull.
9. The marine vessel according to claim 6, wherein the auxiliary windshield is one of a starboard

auxiliary windshield and a port auxiliary windshield, and wherein movement of each of the starboard auxiliary windshield and the port auxiliary windshield into the deployed position provides a weathertight barrier extending from the starboard side to the port side.

10. The marine vessel according to claim 9, wherein the gap is one of a starboard gap and a port gap, and wherein in the deployed position, the starboard auxiliary windshield and the port auxiliary windshield span an upper portion of the starboard gap and the port gap, respectively, and further comprising a starboard wind door and a port wind door which are each movable into and between a retracted position and an extended position in which the starboard wind door and the port wind doors span a lower portion of the starboard gap and the port gap, respectively, so that together the console, the starboard auxiliary windshield, the port auxiliary windshield, the starboard wind door, and the port wind door provide a weathertight barrier extending from the starboard side to the port side and from a deck on the hull to the cover.

11. The marine vessel according to claim 1, further comprising a pivot assembly that pivotably couples an upper end of the auxiliary windshield to the cover along a pivot axis.

12. The marine vessel according to claim 11, wherein the pivot assembly includes an upper actuator configured to pivot the auxiliary windshield into the stowed position and into the deployed position.

13. The marine vessel according to claim 12, wherein the upper actuator includes a linear actuator.

14. The marine vessel according to claim 13, wherein the upper actuator is coupled to the auxiliary windshield by a rocker arm.

15. The marine vessel according to claim 13, wherein the pivot assembly includes a pivot bracket that is fixedly coupled to the auxiliary windshield and wherein extension and retraction of the linear actuator pivots the pivot bracket and the auxiliary windshield about the pivot axis.

16. The marine vessel according to claim 15, wherein the cover includes a window frame and wherein the pivot bracket includes a curved outer surface that remains adjacent to the window frame as the auxiliary windshield is pivoted about the pivot axis into and between the stowed position and the deployed position.

17. The marine vessel according to claim 16, further comprising a wiper seal on the window frame, wherein the curved outer surface slides along the wiper seal as the auxiliary windshield is pivoted about the pivot axis.

18. The marine vessel according to claim 12, further comprising a lower actuator that couples a lower end of the auxiliary windshield to the console and to the hull in the deployed position.

19. The marine vessel according to claim 18, wherein the lower actuator includes a linear actuator that is retained in the lower end of the auxiliary windshield, the lower actuator being movable into an active position which engages the lower end of the auxiliary windshield with the console and with the hull, and an inactive position which disengages the lower end of the auxiliary windshield from the console and the hull.

20. A convertible windshield apparatus for a marine vessel, the convertible windshield apparatus comprising: a cover having a recess, an auxiliary windshield being pivotable into and between a stowed position in which the auxiliary windshield is nested in the recess and a deployed position in which the auxiliary windshield is located at least partially out of the recess, and a pivot assembly that pivotably couples the auxiliary windshield to the cover along a pivot axis, wherein the pivot assembly includes an actuator and a pivot bracket that is fixedly coupled to the auxiliary windshield, and wherein the actuator pivots the pivot bracket and auxiliary windshield about the pivot axis.

21. The convertible windshield apparatus according to claim 20, further comprising a rocker arm that pivotably couples the actuator to the pivot bracket.

22. The convertible windshield apparatus according to claim 21, wherein the actuator includes a linear actuator.

23. The convertible windshield apparatus according to claim 21, wherein the cover includes a

window frame and wherein the pivot bracket has a curved outer surface that remains adjacent to the window frame as the auxiliary windshield is pivoted about the pivot axis into and between the stowed position and the deployed position.

24. The convertible windshield apparatus according to claim 21, wherein the actuator is an upper actuator and further comprising a lower actuator for coupling a lower end of the auxiliary windshield to the marine vessel in the deployed position.

25. A marine vessel comprising: a hull having a starboard side and a port side; a console having a helm and a console windshield that shields the helm from wind; a cover extending over the helm; a starboard auxiliary windshield coupled to the cover, the starboard auxiliary windshield being movable relative to the cover into a stowed position in which the starboard auxiliary windshield is stowed in the cover and into a deployed position in which the starboard auxiliary windshield extends downwardly from the cover into a starboard gap between the console and the starboard side, wherein in the deployed position the starboard auxiliary windshield spans an upper portion of the starboard gap; and a port auxiliary windshield coupled to the cover, the port auxiliary windshield being movable relative to the cover into a stowed position in which the port auxiliary windshield is stowed in the cover and into a deployed position in which the port auxiliary windshield extends downwardly from the cover into a port gap between the console and the port side, wherein in the deployed position the port auxiliary windshield spans an upper portion of the port gap, wherein together with the console windshield, the starboard auxiliary windshield and the port auxiliary windshield provide a weathertight barrier extending from the starboard side to the port side.

26. The marine vessel according to claim 25, further comprising a starboard wind door and a port wind door that are each movable into and between a retracted position and an extended position in which the starboard wind door and the port wind door span a lower portion of the starboard gap and the port gap, respectively, so that together the console, the starboard auxiliary windshield, the port auxiliary windshield, the starboard wind door, and the port wind door provide a weathertight barrier extending from the starboard side to the port side and from a deck on the hull to the cover.
